

LDscnR — LD-aware outlier regions

Testing loci instead of markers, so that the reported false-discovery rate is a statement about the thing you care about.

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github.com/genomeinsights/LDscnR

THE PREMISE

A genome scan tests markers; the object of interest is a locus. Benjamini–Hochberg controls the false discovery proportion *among the units tested*, so a single-SNP scan controls it among SNPs while being judged against a locus-level truth. On one simulated panel, 2,467 significant SNPs were **132 distinct LD clusters**, with the rejected set 6.3× more redundant than the genome as a whole. “5% FDR over 2,467 discoveries” is a guarantee about a different object from the one being counted.

There is a second reason, and it is the one the method exists for. Where the environment is collinear with population structure, residual confounding survives a mixed model and produces *isolated* single-marker hits, while real adaptive loci produce *clustered* signal. **That difference is measured, not assumed:** across 2,547 simulated discoveries with known causal variants, true positives sit in clusters about twice the size and false positives are 3–4× more likely to be singletons.

THE PIPELINE

00 LD-decay estimation Fit the decay of r^2 with distance per chromosome. Every downstream threshold is <i>derived</i> from this fit rather than chosen, so a setting means the same thing on chromosomes with different recombination.	01 Kinship Build the relatedness matrix from LD- <i>independent</i> markers, so it cannot absorb the very regions under selection.
02 Complexity reduction Two stages. Group markers into LD clusters, then merge clusters that are linked and physically contiguous. Grouping is by <i>linkage</i> , not physical distance, so an inversion stays one unit while an adjacent looser block separates. The partition never sees a phenotype.	03 Association scan EMMAX, LFMM, or any test giving one p -value per marker. The method consumes the vector and is agnostic to its source; EMMAX and LFMM give the same conclusions.
04 Cluster summary Simes across a cluster’s members. For methods that emit no per-marker p -value — BayPass BF, XtX — the test is run instead on the cluster’s consensus genotype, at a measured cost in power.	05 Correction over loci BH across clusters. This is the whole point: the correction and the unit now refer to the same object, and no filtering or thresholding is involved.
06 Null diagnostic Read σ^2_e from REML <i>before</i> running any permutation. Near zero, the surrogate null is anti-conservative and can only be read one-sidedly. It fires without needing to know why.	07 Report Loci with a locus-level FDR, and an explicit statement of what the accompanying null can and cannot certify.

WHAT IS DIFFERENT ABOUT IT

Units, not filters

Every filter we tried was rejected on its own evidence — four built on cluster geometry, and a minimum-size floor swept from 1 to 12. The recall lost always exceeded the precision gained. The advantage comes from *changing what is tested*, not from removing anything.

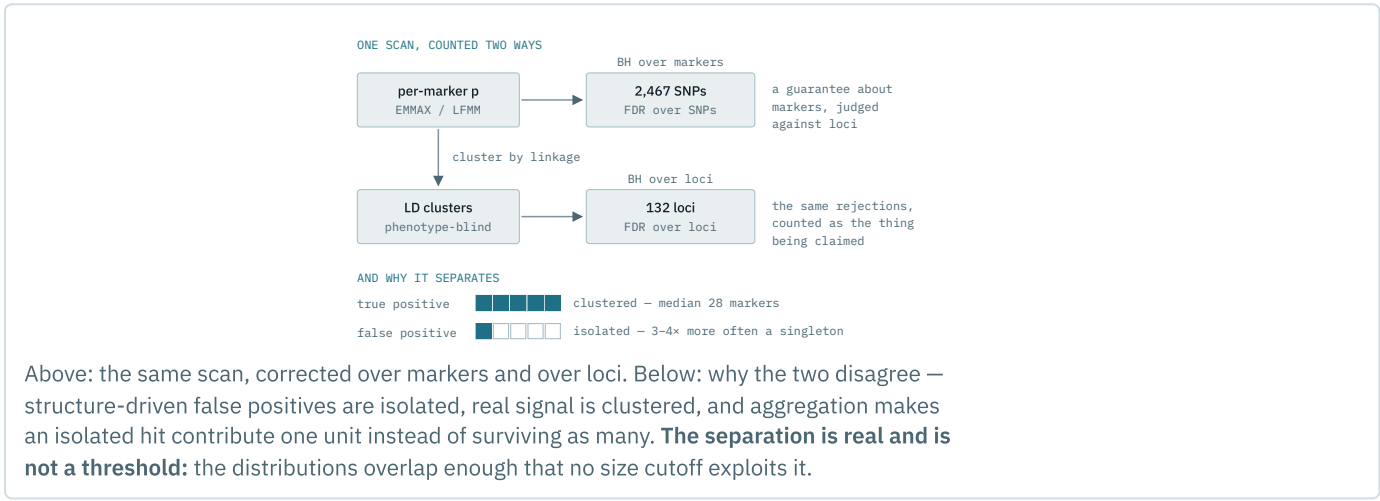
Thresholds derived, not chosen

Merge distance and correlation floors come from each chromosome’s own decay fit. Where a constant remains, it is named as one. A full factorial over the two clustering parameters and the size floor moves the size of the advantage and never its sign.

Honest about the null

In the regime this method targets, surrogate nulls are anti-conservative — measured against simulations with known truth. A negative result is therefore trustworthy; survivors are *candidates*, pending independent evidence. We report the asymmetry rather than a number we cannot support.

THE IDEA IN ONE PICTURE



HOW IT DIFFERS FROM EXISTING APPROACHES

Haplotype and LD-pattern sweep scans [H12](#) · [OmegaPlus](#)

These read elevated LD as direct evidence of selection. LDscnR does not — elevated LD also arises from inversions, recombination-rate variation, demography and structure. Local LD is used only to define *units*; it is never interpreted as a test of selection and never folded into the evidence.

LD-based complexity reduction [the LDna lineage](#)

Complexity reduction originates here, not with LDscnR. Fang *et al.* (2021) shrank the test count so the permutation of Li *et al.* (2018) became affordable, and scored regions by the proportion of settings finding them significant. LDscnR keeps the reduction and drops that consistency score as an inferential device: swept over our grid it is informative only at its ceiling — units called under *every* setting are 4x enriched for lying near a causal variant, everything below flat. It is an indicator, not a ranking. What is new is that the reduction is **benchmarked against simulations with known causal variants**, with a convention-free false-positive control.

Window-based regional aggregation [WZA](#)

Fixed physical windows assume adjacent variants carry one evolutionary signal; recombination means they often do not, so aggregation dilutes highly localised signal. LDscnR takes units from the topology of the local LD network — overlapping but distinct signals in the same interval stay distinct — and the joining thresholds self-calibrate to each chromosome’s LD decay.

How significance is established — and its limit

Permutation is the acknowledged gold standard, because it accounts for the genome’s correlation structure directly. Under population structure the standard version is invalid and the correct substitute is specified: surrogate phenotypes from $MVN(0, \sigma_g^2 A + \sigma_e^2 I)$. **We find it fails in exactly the regime it is needed.** Where the covariate is collinear with structure, REML returns $\sigma_e^2 \approx 0$ and the surrogate null runs several-fold too light in the tail where BH operates — anti-conservative in 8 of 9 simulated panels. The failure is detectable in advance from the variance split and its direction is known, which is what makes it usable: a negative result can be trusted, survivors are candidates needing independent corroboration. We regard this as a finding about structure-aware nulls, not a limitation of this pipeline.

STATUS

Benchmarked on forward-in-time, individual-based simulations (NEMO) with known causal variants: 80 panels across four selection regimes and two background-selection arms, with half of all markers on chromosomes carrying no causal variant, so that a discovery there is a false positive by construction and no scoring convention is involved. Cluster-level testing roughly doubles precision (0.35 against 0.19), and the advantage holds in every cell of a full factorial over the clustering parameters and under both association engines. The trade against recall is explicit: the crossover in F_{β} sits at $\beta \approx 2$.

Application to the three-spined stickleback marine–freshwater panel of Fang *et al.* (2021) is in progress, where a single-SNP scan returns three isolated loci matching no known adaptation peak. Corroboration of the cluster-level regions against published recurrent marine–freshwater peaks is being tested against a positional null and is **not yet complete**.

Implemented as an R package: github.com/genomeinsights/LDscnR. Manuscript in preparation — comments welcome at petri@genomeinsights.fi.